

Ava J. Romano

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EDUCATION

California Polytechnic State University - San Luis Obispo

Sept. 2022 – May 2027

Bachelor of Science in Computer Science | Bachelor of Science in Geography and Anthropology

- Relevant coursework: Intro and Advanced GIS, Remote Sensing, Biogeography Methods, Data Structures, Object-Oriented Programming, Software Engineering, Algorithm Design, Database Management

PROJECTS

Northern Spotted Owl Habitat Assessment | *GDAL, QGIS, ArcPro, MaxEnt, Conda, Python*

- Pre-processed environmental and terrain raster variables using the GDAL tool suite in a custom, command-line Conda environment for maximized function customizability, efficiency, and ease of use
- Ran 10 iterations of the Maximum Entropy model in ArcPro with different input variable combinations to enhance accuracy of the resultant habitat suitability map and to test alternate climatic scenarios; Visualized results in QGIS

GeoTIFF Generator | *Python, Python PDAL, Git, VS Code, GeoPandas, PathLib, ArgParse*

- Designed and wrote a Python-based geospatial processing pipeline with Python PDAL and GeoPandas to automate generation of a Canopy Height Model, Digital Elevation Model, and Digital Surface Model from a point-cloud LAS file in order to streamline preprocessing for PyCrown, an existing open-source tree canopy segmentation tool

EXPERIENCE

Undergraduate Research Assistant | *Cal Poly Geoinformatics Lab*

January 2026 – Present

Soil Burn Severity AI-Powered Assessment Map

- Utilized Google Earth Engine to aid in development and refinement of a machine learning model which uses optical remote sensing indices, synthetic aperture radar, and environmental, weather, and terrain covariates to predict the soil burn severity of given geographic areas in the wake of wildfire events in order to inform landslide risk

DamageMap: A Post-Wildfire Damage Assessment Tool

- Aided in iterative software development, model tuning, and application deployment of a computer vision machine learning model which employs satellite imagery to identify structures damaged in wildfire events

Remote Sensing TA | *Cal Poly Geography Department*

August 2026 – Present

- Assisted students with finding and processing satellite data in ArcPro, answered questions, and advised class projects

Tech Lead | *Cal Poly Hack4Impact*

September 2026 – Present

- Managed a team of software developers, held weekly check-in meetings, organized sprints, planned a long-term project schedule, performed code review, and fostered a welcoming and supportive team dynamic
- Worked with product managers, UX designers, and nonprofit clients to map their needs to actionable development tasks

AWS Application Developer | *California Cybersecurity Institute, Cal Poly*

October 2025 – June 2026

- Used the AWS Cloud Development Kit and React to contribute to full-stack development of a gamified cybersecurity education platform intended to provide middle and high-school students with an engaging educational experience
- Iteratively designed educational modules according to UX/UI best practices and user testing feedback
- Supported director with project planning, organization and analysis of development tasks, and team communication

GIS Intern | *AppliedEarthworks*

June 2025 – September 2025

- Georeferenced and digitized historic maps to record ethnographic villages in an ArcGIS feature layer, creating a spatial dataset for heritage preservation and research

Usability and Integration eLearning Assistant | *Cal Poly Office of Student Research*

January 2025 – June 2025

- Developed and refined online learning modules with UX test results to increase usability and enhance learning outcomes

TECHNICAL SKILLS

Languages and Frameworks: Python, C, TypeScript, React, HTML/CSS, SQL

Tools: ArcGIS, QGIS, GDAL, MaxEnt, Git, Conda, AWS, MongoDB, UNIX shell scripting

Concepts: Software Development Lifecycle, CI/CD, MVC Architecture, Agile development, REST API, HTTP/IP Protocols